



ASSIGNMENT COVER SHEET

Course Code and Description	UECS1413 DATABASE SYSTEM FUNDAMENTALS		
Lecturer Name	Dr.Sugumaran a/l Nallusamy		
Assignment Title	Assignment 3		
DECLARATION			
We declare that this is a group assignment and that no part of this submission has been copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part been written for us by another person.			
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Description	Full Marks	Marks Obtained
Tables	10	
Information	30	
Total	40	

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SQL Script to create tables with sample data

DROP TABLE PurchaseDetail CASCADE CONSTRAINTS;

DROP TABLE OrderDetail CASCADE CONSTRAINTS;

DROP TABLE Delivery CASCADE CONSTRAINTS;

DROP TABLE Orders CASCADE CONSTRAINTS;

DROP TABLE Purchase CASCADE CONSTRAINTS;

DROP TABLE Product CASCADE CONSTRAINTS;

DROP TABLE Vendor CASCADE CONSTRAINTS;

DROP TABLE Customer CASCADE CONSTRAINTS;

DROP TABLE Employee CASCADE CONSTRAINTS;

-- Customer Table

CREATE TABLE Customer (

 Cust_ID VARCHAR2(8) NOT NULL,

 Cust_Type VARCHAR2(20) CHECK (Cust_Type IN ('customer', 'agent', 'dropshipper')),

 Cust_Name VARCHAR2(25) NOT NULL,

 Cust_Email VARCHAR2(24),

 Cust_Contact VARCHAR2(15),

 Cust_Add VARCHAR2(30),

 CONSTRAINT customer_cust_id_pk PRIMARY KEY (Cust_ID));

-- Employee Table

CREATE TABLE Employee (

 Emp_ID VARCHAR2(8) NOT NULL,

 Emp_Name VARCHAR2(30) NOT NULL,

 Emp_Contact VARCHAR2(15),

 CONSTRAINT employee_emp_id_pk PRIMARY KEY (Emp_ID));

-- Orders Table

```
CREATE TABLE Orders (  
    Ord_ID VARCHAR2(8) NOT NULL,  
    Cust_ID VARCHAR2(8) NOT NULL,  
    Ord_Date DATE DEFAULT SYSDATE,  
    Ord_CommRate NUMBER(5,2),  
    OD_PayMethod VARCHAR2(20) CHECK (OD_PayMethod IN ('CreditCard', 'TnG',  
'BankTransfer')),  
    CONSTRAINT orders_ord_id_pk PRIMARY KEY (Ord_ID),  
    CONSTRAINT orders_cust_id_fk FOREIGN KEY (Cust_ID) REFERENCES  
Customer(Cust_ID));
```

-- Delivery Table

```
CREATE TABLE Delivery (  
    Del_ID VARCHAR2(10) NOT NULL,  
    Ord_ID VARCHAR2(8) NOT NULL,  
    Emp_ID VARCHAR2(8) NOT NULL,  
    Del_Fee NUMBER(5,2) DEFAULT 0,  
    Del_Method VARCHAR2(20),  
    CONSTRAINT delivery_del_id_pk PRIMARY KEY (Del_ID),  
    CONSTRAINT delivery_ord_id_fk FOREIGN KEY (Ord_ID) REFERENCES  
Orders(Ord_ID),  
    CONSTRAINT delivery_emp_id_fk FOREIGN KEY (Emp_ID) REFERENCES  
Employee(Emp_ID));
```

-- Product Table

```
CREATE TABLE Product (  
    Prod_ID VARCHAR2(15) NOT NULL,  
    Prod_Type VARCHAR2(30) CHECK (Prod_Type IN ('Sharifah Ready to Eat Products',  
'Biodegradable Packaging Items')),  
    Prod_SubCategory VARCHAR2(16),  
    Prod_Info VARCHAR2(40),  
    Prod_Qty NUMBER(10) DEFAULT 0,  
    Prod_QtyPerUnit NUMBER(10),  
    Prod_Price NUMBER(8,2) DEFAULT 0,  
    Prod_Weight NUMBER(10,2) DEFAULT 0,  
    CONSTRAINT product_prod_id_pk PRIMARY KEY (Prod_ID));
```

-- OrderDetail Table

```
CREATE TABLE OrderDetail (  
    OD_ID VARCHAR2(10) NOT NULL,  
    Ord_ID VARCHAR2(8) NOT NULL,  
    Prod_ID VARCHAR2(15) NOT NULL,  
    OD_Qty NUMBER(8) DEFAULT 1,  
    O_Disc NUMBER(5,2) DEFAULT 0,  
    Ord_Price NUMBER(7,2) DEFAULT 0,  
    CONSTRAINT orderdetail_od_id_pk PRIMARY KEY (OD_ID),  
    CONSTRAINT orderdetail_ord_id_fk FOREIGN KEY (Ord_ID) REFERENCES  
Orders(Ord_ID),  
    CONSTRAINT orderdetail_prod_id_fk FOREIGN KEY (Prod_ID) REFERENCES  
Product(Prod_ID));
```

-- Vendor Table

```
CREATE TABLE Vendor (  
    Vend_ID VARCHAR2(8) NOT NULL,  
    Vend_Category VARCHAR2(21) CHECK (Vend_Category IN ('Sharifah Food Sdn Bhd',  
'Other vendors')),  
    Vend_Name VARCHAR2(30) NOT NULL,  
    Vend_Contact VARCHAR2(15),  
    Vend_BD NUMBER(10),  
    CONSTRAINT vendor_vend_id_pk PRIMARY KEY (Vend_ID));
```

-- Purchase Table

```
CREATE TABLE Purchase (  
    P_ID VARCHAR2(8) NOT NULL,  
    Vend_ID VARCHAR2(5) NOT NULL,  
    P_Date DATE DEFAULT SYSDATE,  
    CONSTRAINT purchase_p_id_pk PRIMARY KEY (P_ID),  
    CONSTRAINT purchase_vend_id_fk FOREIGN KEY (Vend_ID) REFERENCES  
Vendor(Vend_ID));
```

-- PurchaseDetail Table

```
CREATE TABLE PurchaseDetail (  
    P_ID VARCHAR2(8) NOT NULL,  
    Prod_ID VARCHAR2(15) NOT NULL,  
    P_Qty NUMBER(8) DEFAULT 0,  
    P_Rec NUMBER(7,2) DEFAULT 0,  
    CONSTRAINT purchasedetail_pk PRIMARY KEY (P_ID, Prod_ID),  
    CONSTRAINT purchasedetail_p_id_fk FOREIGN KEY (P_ID) REFERENCES  
Purchase(P_ID),  
    CONSTRAINT purchasedetail_prod_id_fk FOREIGN KEY (Prod_ID) REFERENCES  
Product(Prod_ID));
```

-----Customer Table Insert-----

-- Regular Customers

```
INSERT INTO Customer (Cust_ID, Cust_Type, Cust_Name, Cust_Email, Cust_Contact, Cust_Add) VALUES
```

```
('C001', 'customer', 'Ali Baba', 'alibaba@gmail.com', '0123456789', '10 Jalan Bunga Raya, KL'),
```

```
('C002', 'customer', 'Siti Liza', 'sitilizzo@gmail.com', '0172345678', '21 Jalan Teratai, Seremban');
```

-- Agents

```
INSERT INTO Customer (Cust_ID, Cust_Type, Cust_Name, Cust_Email, Cust_Contact, Cust_Add) VALUES
```

```
('A001', 'agent', 'Nurul Aini', 'nurulaini@gmail.com', '0134567890', '88 Jalan Damai, Shah Alam'),
```

```
('A002', 'agent', 'Faiz Saduk', 'faizsaduk@gmail.com', '0168765432', '31 Jalan Ria, Ipoh');
```

-- Dropshippers

```
INSERT INTO Customer (Cust_ID, Cust_Type, Cust_Name, Cust_Email, Cust_Contact, Cust_Add) VALUES
```

```
('D001', 'dropshipper', 'Jason Derulo', 'jasonderulo@gmail.com', '0145678901', '55 Lorong Anggerik, Penang'),
```

```
('D002', 'dropshipper', 'Alicia Keys', 'aliciakeys@gmail.com', '0189988776', '19 Taman Bukit, Johor Bahru');
```

-----Employee Table Insert-----

```
INSERT INTO Employee (Emp_ID, Emp_Name, Emp_Contact)
```

```
VALUES ('E001', 'Thomas Tan', '0112233445');
```


-----Orders & OrderDetails Table Insert-----

-- Order for Regular Customer (Ali) - Sharifah Ready to Eat Products

-- No discount (Ord_CommRate = 0)

INSERT INTO Orders (Ord_ID, Ord_Date, Ord_CommRate, OD_PayMethod, Cust_ID)
VALUES ('O001', TO_DATE('01-05-2025', 'DD-MM-YYYY'), 0, 'CreditCard', 'C001');

-- PR001: 2 units @ 13.90 = 27.80

-- PR003: 1 unit @ 23.90 = 23.90

INSERT INTO OrderDetail (OD_ID, OD_Qty, O_Disc, Ord_Price, Ord_ID, Prod_ID)
VALUES

('OD001', 2, 0, 27.80, 'O001', 'PR001'),

('OD002', 1, 0, 23.90, 'O001', 'PR003');

UPDATE Product SET Prod_Qty = Prod_Qty - 2 WHERE Prod_ID = 'PR001';

UPDATE Product SET Prod_Qty = Prod_Qty - 1 WHERE Prod_ID = 'PR003';

-- Order for Regular Customer (Siti) - Biodegradable Packaging Items

-- No discount

INSERT INTO Orders (Ord_ID, Ord_Date, Ord_CommRate, OD_PayMethod, Cust_ID)
VALUES ('O002', TO_DATE('02-05-2025', 'DD-MM-YYYY'), 0, 'BankTransfer', 'C002');

-- HIT-CTR-001: 3 units @ 285.00 = 855.00

-- HIT-CB-001: 2 units @ 7.50 = 15.00

```
INSERT INTO OrderDetail (OD_ID, OD_Qty, O_Disc, Ord_Price, Ord_ID, Prod_ID)
VALUES
```

```
('OD003', 3, 0, 855.00, 'O002', 'HIT-CTR-001'),
```

```
('OD004', 2, 0, 15.00, 'O002', 'HIT-CB-001');
```

```
UPDATE Product SET Prod_Qty = Prod_Qty - 3 WHERE Prod_ID = 'HIT-CTR-001';
```

```
UPDATE Product SET Prod_Qty = Prod_Qty - 2 WHERE Prod_ID = 'HIT-CB-001';
```

```
-----
-- Order for Agent (Aini) - Sharifah Ready to Eat Products
```

```
-- 10% discount on each item
-----
```

```
INSERT INTO Orders (Ord_ID, Ord_Date, Ord_CommRate, OD_PayMethod, Cust_ID)
```

```
VALUES ('O003', TO_DATE('03-05-2025', 'DD-MM-YYYY'), 10, 'BankTransfer', 'A001');
```

```
-- PR002: 10 units @ 12.90 = 129.00 - 10% = 116.10
```

```
-- PR005: 50 units @ 11.90 = 595.00 - 10% = 535.5
```

```
INSERT INTO OrderDetail (OD_ID, OD_Qty, O_Disc, Ord_Price, Ord_ID, Prod_ID)
VALUES
```

```
('OD005', 10, 12.90, 116.10, 'O003', 'PR002'),
```

```
('OD006', 50, 59.50, 535.5, 'O003', 'PR005');
```

```
UPDATE Product SET Prod_Qty = Prod_Qty - 10 WHERE Prod_ID = 'PR002';
```

```
UPDATE Product SET Prod_Qty = Prod_Qty - 50 WHERE Prod_ID = 'PR005';
```

```
-----
-- Order for Agent (Faiz) - Biodegradable Packaging Items
```

```
-- 3% discount
-----
```

```
INSERT INTO Orders (Ord_ID, Ord_Date, Ord_CommRate, OD_PayMethod, Cust_ID)
VALUES ('O004', TO_DATE('04-05-2025', 'DD-MM-YYYY'), 3, 'TnG', 'A002');
```

```
-- HIT-CTR-005: 20 units @ 195.00 = 3900.00 - 3% = 3783.00
```

```
-- HIT-UTS-003: 100 units @ 180.00 = 18000.00 - 3% = 17460.00
```

```
INSERT INTO OrderDetail (OD_ID, OD_Qty, O_Disc, Ord_Price, Ord_ID, Prod_ID)
VALUES
```

```
('OD007', 20, 117.00, 3783.00, 'O004', 'HIT-CTR-005'),
```

```
('OD008', 100, 540.00, 17460.00, 'O004', 'HIT-UTS-003');
```

```
UPDATE Product SET Prod_Qty = Prod_Qty - 20 WHERE Prod_ID = 'HIT-CTR-005';
```

```
UPDATE Product SET Prod_Qty = Prod_Qty - 100 WHERE Prod_ID = 'HIT-UTS-003';
```

```
-----
-- Order for Dropshipper (Jason) - Sharifah Ready to Eat Products
```

```
-- 10% discount
-----
```

```
INSERT INTO Orders (Ord_ID, Ord_Date, Ord_CommRate, OD_PayMethod, Cust_ID)
VALUES ('O005', TO_DATE('05-05-2025', 'DD-MM-YYYY'), 10, 'CreditCard', 'D001');
```

```
-- PR001: 15 units @ 13.90 = 208.50 - 10% = 187.65
```

```
-- PR006: 10 units @ 11.90 = 119.00 - 10% = 107.10
```

```
INSERT INTO OrderDetail (OD_ID, OD_Qty, O_Disc, Ord_Price, Ord_ID, Prod_ID)
VALUES
```

```
('OD009', 15, 20.85, 187.65, 'O005', 'PR001'),
```

```
('OD010', 10, 11.90, 107.10, 'O005', 'PR006');
```

```
UPDATE Product SET Prod_Qty = Prod_Qty - 15 WHERE Prod_ID = 'PR001';
```

```
UPDATE Product SET Prod_Qty = Prod_Qty - 10 WHERE Prod_ID = 'PR006';
```

-- Order for Dropshipper (Alicia) - Biodegradable Packaging Items

-- 3% discount

```
INSERT INTO Orders (Ord_ID, Ord_Date, Ord_CommRate, OD_PayMethod, Cust_ID)
VALUES ('O006', TO_DATE('06-05-2025', 'DD-MM-YYYY'), 3, 'BankTransfer', 'D002');
```

-- HIT-CTR-002: 30 units @ 225.00 = 6750.00 - 3% = 6547.50

-- HIT-CB-005: 20 units @ 23.50 = 470.00 - 3% = 455.90

```
INSERT INTO OrderDetail (OD_ID, OD_Qty, O_Disc, Ord_Price, Ord_ID, Prod_ID)
VALUES
```

('OD011', 30, 202.50, 6547.50, 'O006', 'HIT-CTR-002'),

('OD012', 20, 14.10, 455.90, 'O006', 'HIT-CB-005');

```
UPDATE Product SET Prod_Qty = Prod_Qty - 30 WHERE Prod_ID = 'HIT-CTR-002';
```

```
UPDATE Product SET Prod_Qty = Prod_Qty - 20 WHERE Prod_ID = 'HIT-CB-005';
```

----Delivery Table Insert-----

-- Delivery for Order O001 (Ali Ahmad) - HIT's in-house service

```
INSERT INTO Delivery (Del_ID, Del_Fee, Del_Method, Ord_ID, Emp_ID)
VALUES ('D001', 15.00, 'HIT In-house Service', 'O001', 'E001');
```

-- Delivery for Order O002 (Siti Aminah) - Grab

```
INSERT INTO Delivery (Del_ID, Del_Fee, Del_Method, Ord_ID, Emp_ID)
VALUES ('D002', 20.00, 'Grab', 'O002', 'E001');
```

-- Delivery for Order O003 (Nurul Aini) - Lalamove

```
INSERT INTO Delivery (Del_ID, Del_Fee, Del_Method, Ord_ID, Emp_ID)
VALUES ('D003', 25.00, 'Lalamove', 'O003', 'E001');
```

-- Delivery for Order O004 (Mohd Faiz) - HIT's in-house service

```
INSERT INTO Delivery (Del_ID, Del_Fee, Del_Method, Ord_ID, Emp_ID)
VALUES ('D004', 18.00, 'HIT In-house Service', 'O004', 'E001');
```

-- Delivery for Order O005 (Jason Lim) - Grab

```
INSERT INTO Delivery (Del_ID, Del_Fee, Del_Method, Ord_ID, Emp_ID)
VALUES ('D005', 12.00, 'Grab', 'O005', 'E001');
```

-- Delivery for Order O006 (Alicia Wong) - Lalamove

```
INSERT INTO Delivery (Del_ID, Del_Fee, Del_Method, Ord_ID, Emp_ID)
VALUES ('D006', 10.00, 'Lalamove', 'O006', 'E001');
```

-- Product Table Insert--

-- Sharifah Ready to Eat Products --

-- Ready to Eat Products

INSERT INTO Product (Prod_ID, Prod_Type, Prod_Info, Prod_SubCategory, Prod_Qty, Prod_QtyPerUnit, Prod_Price, Prod_Weight) VALUES

('PR001', 'Sharifah Ready to Eat Products', 'NASI BRIYANI GAM AYAM', 'Ready to Eat', 500, 1, 13.90, 250),

('PR002', 'Sharifah Ready to Eat Products', 'SAMBAL SOTONG KERING', 'Ready to Eat', 500, 1, 12.90, 180),

('PR003', 'Sharifah Ready to Eat Products', 'RENDANG DENDENG (200G)', 'Ready to Eat', 500, 1, 23.90, 240),

('PR004', 'Sharifah Ready to Eat Products', 'SAMBAL TUMIS IKAN BILIS', 'Ready to Eat', 500, 1, 6.90, 140),

('PR005', 'Sharifah Ready to Eat Products', 'NASI GORENG KAMPUNG', 'Ready to Eat', 500, 1, 11.90, 250),

('PR006', 'Sharifah Ready to Eat Products', 'NASI GORENG UDANG', 'Ready to Eat', 500, 1, 11.90, 250),

('PR007', 'Sharifah Ready to Eat Products', 'NASI GORENG IKAN MASIN', 'Ready to Eat', 500, 1, 11.90, 250),

('PR008', 'Sharifah Ready to Eat Products', 'NASI GORENG AYAM', 'Ready to Eat', 500, 1, 11.90, 250),

('PR009', 'Sharifah Ready to Eat Products', 'NASI GORENG DAGING', 'Ready to Eat', 500, 1, 11.90, 250),

('PR010', 'Sharifah Ready to Eat Products', 'KARI AYAM', 'Ready to Eat', 500, 1, 8.90, 140),

('PR011', 'Sharifah Ready to Eat Products', 'NASI PUTIH', 'Ready to Eat', 500, 1, 4.50, 250),

('PR012', 'Sharifah Ready to Eat Products', 'RENDANG DENDENG MINI (120G)', 'Ready to Eat', 500, 1, 14.90, 120);

-- Ready to Cook Products

INSERT INTO Product (Prod_ID, Prod_Type, Prod_Info, Prod_SubCategory, Prod_Qty, Prod_QtyPerUnit, Prod_Price, Prod_Weight) VALUES

('PR013', 'Sharifah Ready to Eat Products', 'NASI BRIYANI GAM 535G', 'Ready to Cook', 500, 1, 21.90, 535),

('PR014', 'Sharifah Ready to Eat Products', 'NASI BRIYANI HERBA 395G', 'Ready to Cook', 500, 1, 21.90, 395),

('PR015', 'Sharifah Ready to Eat Products', 'NASI BRIYANI BUKHARI 500G', 'Ready to Cook', 500, 1, 21.90, 500);

-- Paste Products

INSERT INTO Product (Prod_ID, Prod_Type, Prod_Info, Prod_SubCategory, Prod_Qty, Prod_QtyPerUnit, Prod_Price, Prod_Weight) VALUES

('PR016', 'Sharifah Ready to Eat Products', 'PES KARI KEPALA IKAN 120G', 'Paste', 500, 1, 5.90, 120),

('PR017', 'Sharifah Ready to Eat Products', 'PES NASI GORENG KAMPUNG 120G', 'Paste', 500, 1, 5.90, 120),

('PR018', 'Sharifah Ready to Eat Products', 'PES NASI BRIYANI GAM 120G', 'Paste', 500, 1, 5.90, 120),

('PR019', 'Sharifah Ready to Eat Products', 'PES MASAK KERUTUK 120G', 'Paste', 500, 1, 5.90, 120),

('PR020', 'Sharifah Ready to Eat Products', 'PES ASAM PEDAS 120G', 'Paste', 500, 1, 5.90, 120),

('PR021', 'Sharifah Ready to Eat Products', 'PES SAMBAL TUMIS 120G', 'Paste', 500, 1, 5.90, 120),

('PR022', 'Sharifah Ready to Eat Products', 'PES MASAK TIGA RASA 120G', 'Paste', 500, 1, 5.90, 120),

('PR023', 'Sharifah Ready to Eat Products', 'PES MIE GORENG 120G', 'Paste', 500, 1, 5.90, 120);

-- Biodegradable Packaging Items --

-- Carry Bag

INSERT INTO Product (Prod_ID, Prod_Type, Prod_Info, Prod_SubCategory, Prod_Qty, Prod_QtyPerUnit, Prod_Price, Prod_Weight) VALUES

('HIT-CB-001', 'Biodegradable Packaging Items', 'E20 BIO SINGLET 12x12', 'Carry Bag', 500, 50, 7.50, NULL),

('HIT-CB-002', 'Biodegradable Packaging Items', 'E30 BIO SINGLET 15x15', 'Carry Bag', 500, 50, 11.50, NULL),

('HIT-CB-003', 'Biodegradable Packaging Items', 'E40 BIO SINGLET 17x19', 'Carry Bag', 500, 40, 13.00, NULL),

('HIT-CB-004', 'Biodegradable Packaging Items', 'E48 BIO SINGLET 18x22', 'Carry Bag', 500, 40, 15.50, NULL),

('HIT-CB-005', 'Biodegradable Packaging Items', 'E55 BIO SINGLET 22x24', 'Carry Bag', 500, 40, 23.50, NULL),

('HIT-CB-006', 'Biodegradable Packaging Items', 'E60 BIO SINGLET 24x30', 'Carry Bag', 500, 40, 35.00, NULL),

('HIT-CB-007', 'Biodegradable Packaging Items', 'PAPER BAG WITH BASE SOS NO.4', 'Carry Bag', 500, 1000, 80.00, NULL),

('HIT-CB-008', 'Biodegradable Packaging Items', 'PAPER BAG WITH BASE SOS NO.6', 'Carry Bag', 500, 1000, 90.00, NULL),

('HIT-CB-009', 'Biodegradable Packaging Items', 'PAPER BAG WITH BASE SOS NO.8', 'Carry Bag', 500, 1000, 100.00, NULL);

-- Containers

INSERT INTO Product (Prod_ID, Prod_Type, Prod_Info, Prod_SubCategory, Prod_Qty, Prod_QtyPerUnit, Prod_Price, Prod_Weight) VALUES

('HIT-CTR-001', 'Biodegradable Packaging Items', '2 Compartment Lunch Box (600ml)', 'Container', 500, 400, 285.00, NULL),

('HIT-CTR-002', 'Biodegradable Packaging Items', '4oz corn starch sauce cup', 'Container', 500, 1000, 225.00, NULL),

('HIT-CTR-003', 'Biodegradable Packaging Items', '2oz corn starch sauce cup', 'Container', 500, 1000, 205.00, NULL),

('HIT-CTR-004', 'Biodegradable Packaging Items', 'Corn Starch 800ml bowl', 'Container', 500, 300, 265.00, NULL),

('HIT-CTR-005', 'Biodegradable Packaging Items', 'Corn Starch 400ml bowl', 'Container', 500, 300, 195.00, NULL),

('HIT-CTR-006', 'Biodegradable Packaging Items', 'Corn Starch 450ml bowl', 'Container', 500, 300, 205.00, NULL),

('HIT-CTR-007', 'Biodegradable Packaging Items', 'Corn Starch 280ml bowl', 'Container', 500, 300, 170.00, NULL),

('HIT-CTR-008', 'Biodegradable Packaging Items', '4 compartment lunch box', 'Container', 500, 200, 257.00, NULL),

('HIT-CTR-009', 'Biodegradable Packaging Items', '3 compartment lunch box', 'Container', 500, 200, 250.00, NULL);

-- Cups

INSERT INTO Product (Prod_ID, Prod_Type, Prod_Info, Prod_SubCategory, Prod_Qty, Prod_QtyPerUnit, Prod_Price, Prod_Weight) VALUES

('HIT-CUP-001', 'Biodegradable Packaging Items', '16oz Corn starch Cup', 'Cup', 500, 1000, 320.00, NULL),

('HIT-CUP-002', 'Biodegradable Packaging Items', '16oz Cornstarch Cold Cup Lid', 'Cup', 500, 1000, 150.00, NULL),

('HIT-CUP-003', 'Biodegradable Packaging Items', '90mm paper lid - white', 'Cup', 500, 2000, 235.00, NULL),

('HIT-CUP-004', 'Biodegradable Packaging Items', '90mm PLA lid', 'Cup', 500, 1000, 270.00, NULL),

('HIT-CUP-005', 'Biodegradable Packaging Items', '80 mm paper based lid white', 'Cup', 500, 1000, 225.00, NULL),

('HIT-CUP-006', 'Biodegradable Packaging Items', '12oz Bio Bagasse Cup (Without Lid)', 'Cup', 500, 1000, 605.00, NULL),

('HIT-CUP-007', 'Biodegradable Packaging Items', '12oz RIPPLE WRAP PAPER CUP', 'Cup', 500, 500, 195.00, NULL),

('HIT-CUP-008', 'Biodegradable Packaging Items', '8oz RIPPLE WRAP PAPER CUP', 'Cup', 500, 1000, 145.00, NULL),

('HIT-CUP-009', 'Biodegradable Packaging Items', 'Paper Cup 16oz', 'Cup', 500, 500, 230.00, NULL);

-- Utensil

INSERT INTO Product (Prod_ID, Prod_Type, Prod_Info, Prod_SubCategory, Prod_Qty, Prod_QtyPerUnit, Prod_Price, Prod_Weight) VALUES

('HIT-UTS-001', 'Biodegradable Packaging Items', 'Wooden Stirrer', 'Utensil', 500, 10000, 125.00, NULL),

('HIT-UTS-002', 'Biodegradable Packaging Items', 'Spoon Corn Starch', 'Utensil', 500, 1000, 180.00, NULL),

('HIT-UTS-003', 'Biodegradable Packaging Items', 'Fork Corn Starch', 'Utensil', 500, 1000, 180.00, NULL),

('HIT-UTS-004', 'Biodegradable Packaging Items', 'Wooden Spoon', 'Utensil', 500, 2500, 320.00, NULL),

('HIT-UTS-005', 'Biodegradable Packaging Items', 'Wooden Fork', 'Utensil', 500, 2500, 320.00, NULL),

('HIT-UTS-006', 'Biodegradable Packaging Items', '3 in 1 Cutlery', 'Utensil', 500, 500, 225.00, NULL);

-----Vendor Table Insert-----

--Sharifah Food Sdn Bhd Vendor

```
INSERT INTO Vendor (Vend_ID, Vend_Category, Vend_Name, Vend_Contact, Vend_BD)
VALUES ('V001', 'Sharifah Food Sdn Bhd', 'Sharifah Food Sdn Bhd', '0123456789',
'20230501');
```

-- Other Vendor

```
INSERT INTO Vendor (Vend_ID, Vend_Category, Vend_Name, Vend_Contact, Vend_BD)
VALUES
('V002', 'Other vendors', 'Eco Pack Supplies', '0198765432', '20230415'),
('V003', 'Other vendors', 'Green Packaging Ltd', '0181122334', '20230510'),
('V004', 'Other vendors', 'Fresh Supplies Co.', '0174433221', '20230425');
```

-----Purchase & Purchase Detail Table Insert-----

-- Purchase for Vendor Sharifah Food Sdn Bhd (V001)

INSERT INTO Purchase (P_ID, P_Date, Vend_ID)

VALUES ('P001', TO_DATE('01-05-2025', 'DD-MM-YYYY'), 'V001');

-- Purchase Detail for Purchase P001 (Sharifah Food Sdn Bhd)

INSERT INTO PurchaseDetail (P_ID, Prod_ID, P_Qty, P_Rec) VALUES

('P001', 'PR001', 50, 500.00),

('P001', 'PR002', 50, 300.00),

('P001', 'PR003', 50, 600.00);

-- Update Product quantities for P001 purchase

UPDATE Product

SET Prod_Qty = Prod_Qty + 50

WHERE Prod_ID IN ('PR001', 'PR002', 'PR003');

-- Purchase for Vendor Eco Pack Supplies (V002)

INSERT INTO Purchase (P_ID, P_Date, Vend_ID)

VALUES ('P002', TO_DATE('02-05-2025', 'DD-MM-YYYY'), 'V002');

-- Purchase Detail for Purchase P002 (Eco Pack Supplies)

INSERT INTO PurchaseDetail (P_ID, Prod_ID, P_Qty, P_Rec) VALUES

('P002', 'HIT-CUP-001', 200, 1000.00),

('P002', 'HIT-CTR-001', 200, 1500.00),

('P002', 'HIT-CB-001', 200, 50.00);

-- Update Product quantities for P002 purchase

UPDATE Product

```

SET Prod_Qty = Prod_Qty + 200
WHERE Prod_ID IN ('HIT-CUP-001', 'HIT-CTR-001', 'HIT-CB-001');
-----
-----

-- Purchase for Vendor Green Packaging Ltd (V003)
INSERT INTO Purchase (P_ID, P_Date, Vend_ID)
VALUES ('P003', TO_DATE('03-05-2025', 'DD-MM-YYYY'), 'V003');

-- Purchase Detail for Purchase P003 (Green Packaging Ltd)
INSERT INTO PurchaseDetail (P_ID, Prod_ID, P_Qty, P_Rec) VALUES
('P003', 'HIT-CUP-002', 150, 1200.00),
('P003', 'HIT-CTR-005', 150, 1000.00);

-- Update Product quantities for P003 purchase
UPDATE Product
SET Prod_Qty = Prod_Qty + 150
WHERE Prod_ID IN ('HIT-CUP-002', 'HIT-CTR-005');
-----
-----

-- Purchase for Vendor Fresh Supplies Co. (V004)
INSERT INTO Purchase (P_ID, P_Date, Vend_ID)
VALUES ('P004', TO_DATE('04-05-2025', 'DD-MM-YYYY'), 'V004');

-- Purchase Detail for Purchase P004 (Fresh Supplies Co.)
INSERT INTO PurchaseDetail (P_ID, Prod_ID, P_Qty, P_Rec) VALUES
('P004', 'HIT-UTS-001', 100, 500.00),
('P004', 'HIT-CB-004', 100, 700.00);

-- Update Product quantities for P004 purchase
UPDATE Product

```

```
SET Prod_Qty = Prod_Qty + 100
WHERE Prod_ID IN ('HIT-UTS-001', 'HIT-CB-004');
```

```
-- Formatting Settings--
```

```
SET LINESIZE 250
```

```
SET PAGESIZE 100
```

```
-- Customer Table
```

```
COLUMN CUST_ID      FORMAT A8
```

```
COLUMN CUST_TYPE    FORMAT A14
```

```
COLUMN CUST_NAME     FORMAT A18
```

```
COLUMN CUST_EMAIL    FORMAT A30
```

```
COLUMN CUST_CONTACT  FORMAT A15
```

```
COLUMN CUST_ADD      FORMAT A30
```

```
-- Employee Table
```

```
COLUMN EMP_ID       FORMAT A8
```

```
COLUMN EMP_NAME     FORMAT A30
```

```
COLUMN EMP_CONTACT  FORMAT A15
```

```
-- Orders Table
```

```
COLUMN ORD_ID       FORMAT A8
```

```
COLUMN CUST_ID      FORMAT A8
```

```
COLUMN ORD_DATE     FORMAT A14
```

```
COLUMN ORD_COMMRATE FORMAT 990.99
```

```
COLUMN OD_PAYMETHOD FORMAT A15
```

-- Delivery Table

COLUMN DEL_ID	FORMAT A10
COLUMN ORD_ID	FORMAT A8
COLUMN EMP_ID	FORMAT A8
COLUMN DEL_FEE	FORMAT 9990.99
COLUMN DEL_METHOD	FORMAT A22

-- Product Table

COLUMN PROD_ID	FORMAT A15
COLUMN PROD_TYPE	FORMAT A35
COLUMN PROD_SUBCATEGORY	FORMAT A20
COLUMN PROD_INFO	FORMAT A40
COLUMN PROD_QTY	FORMAT 99999999
COLUMN PROD_QTYPERUNIT	FORMAT 99999999
COLUMN PROD_PRICE	FORMAT 9990.99
COLUMN PROD_WEIGHT	FORMAT 9990.99

-- OrderDetail Table

COLUMN OD_ID	FORMAT A10
COLUMN ORD_ID	FORMAT A8
COLUMN PROD_ID	FORMAT A15
COLUMN OD_QTY	FORMAT 9999
COLUMN O_DISC	FORMAT 990.99
COLUMN ORD_PRICE	FORMAT 9990.99

-- Vendor Table

COLUMN VEND_ID FORMAT A8
COLUMN VEND_CATEGORY FORMAT A25
COLUMN VEND_NAME FORMAT A30
COLUMN VEND_CONTACT FORMAT A15
COLUMN VEND_BD FORMAT 9999999999

-- Purchase Table

COLUMN P_ID FORMAT A8
COLUMN VEND_ID FORMAT A8
COLUMN P_DATE FORMAT A12
COLUMN PURCHASE_ID FORMAT A12
COLUMN PURCHASE_DATE FORMAT A14

-- PurchaseDetail Table

COLUMN P_ID FORMAT A8
COLUMN PROD_ID FORMAT A15
COLUMN P_QTY FORMAT 9999
COLUMN P_REC FORMAT 99990.99

Information retrieval with SQL query

1. Generate Invoice for Vendor

```

SELECT
    p.P_ID AS Purchase_ID,
    v.Vend_Name AS Vendor_Name,
    v.Vend_Contact AS Vendor_Contact,
    TO_CHAR(p.P_Date, 'DD-MM-YYYY') AS Purchase_Date,
    pd.Prod_ID AS Product_ID,
    pr.Prod_Info AS Product_Name,
    pd.P_Qty AS Quantity,
    ROUND(pd.P_Rec, 2) AS Received_Cost,
    NULL AS Total_Invoice
FROM Purchase p
JOIN Vendor v ON p.Vend_ID = v.Vend_ID
JOIN PurchaseDetail pd ON p.P_ID = pd.P_ID
JOIN Product pr ON pd.Prod_ID = pr.Prod_ID
UNION ALL
SELECT
    p.P_ID AS Purchase_ID,
    v.Vend_Name AS Vendor_Name,
    v.Vend_Contact AS Vendor_Contact,
    TO_CHAR(p.P_Date, 'DD-MM-YYYY') AS Purchase_Date,
    NULL AS Product_ID,
    NULL AS Product_Name,
    NULL AS Quantity,
    NULL AS Received_Cost,
    ROUND(SUM(pd.P_Rec), 2) AS Total_Invoice
FROM Purchase p
JOIN Vendor v ON p.Vend_ID = v.Vend_ID
JOIN PurchaseDetail pd ON p.P_ID = pd.P_ID
GROUP BY p.P_ID, v.Vend_Name, v.Vend_Contact, p.P_Date
ORDER BY Purchase_ID, Product_ID NULLS LAST;

```

PURCHASE_ID	VENDOR_NAME	VENDOR_CONTACT	PURCHASE_DATE	PRODUCT_ID	PRODUCT_NAME	QUANTITY	RECEIVED_COST	TOTAL_INVOICE
P001	Sharifah Food Sdn Bhd	0123456789	01-05-2025	PR001	NASI BRIYANI GAM AYAM	50	500	
P001	Sharifah Food Sdn Bhd	0123456789	01-05-2025	PR002	SAMBAL SOTONG KERING	50	300	
P001	Sharifah Food Sdn Bhd	0123456789	01-05-2025	PR003	RENDANG DENDENG (200G)	50	600	
P001	Sharifah Food Sdn Bhd	0123456789	01-05-2025					1400
P002	Eco Pack Supplies	0198765432	02-05-2025	HIT-CB-001	E20 BIO SINGLET 12x12	200	50	
P002	Eco Pack Supplies	0198765432	02-05-2025	HIT-CTR-001	2 Compartment Lunch Box (600ml)	200	1500	
P002	Eco Pack Supplies	0198765432	02-05-2025	HIT-CUP-001	16oz Corn starch Cup	200	1000	
P002	Eco Pack Supplies	0198765432	02-05-2025					2550
P003	Green Packaging Ltd	0181122334	03-05-2025	HIT-CTR-005	Corn Starch 400ml bowl	150	1000	
P003	Green Packaging Ltd	0181122334	03-05-2025	HIT-CUP-002	16oz Cornstarch Cold Cup Lid	150	1200	
P003	Green Packaging Ltd	0181122334	03-05-2025					2200
P004	Fresh Supplies Co.	0174433221	04-05-2025	HIT-CB-004	E48 BIO SINGLET 18x22	100	700	
P004	Fresh Supplies Co.	0174433221	04-05-2025	HIT-UTS-001	Wooden Stirrer	100	500	
P004	Fresh Supplies Co.	0174433221	04-05-2025					1200

Explanation:

This query helps the company generate invoices listing all products purchased from each vendor along with a total invoice value per purchase. This can allow the company

to automate the invoices and filter them. The company can utilize this to better manage their database for invoices with the ability to filter invoices.

2. Generate Invoice for Customer

```
SELECT
    o.Ord_ID AS Order_ID,
    c.Cust_Name,
    c.Cust_Contact,
    TO_CHAR(o.Ord_Date, 'DD-MM-YYYY') AS Order_Date,
    p.Prod_Info AS Product_Name,
    od.OD_Qty AS Quantity,
    ROUND(p.Prod_Price, 2) AS Unit_Price,
    ROUND(od.OD_Qty * p.Prod_Price, 2) AS Original_Price,
    ROUND(od.O_DISC, 2) AS Discount_Applied,
    ROUND(od.OD_Qty * p.Prod_Price - od.O_DISC, 2) AS Total_Price,
    ROUND(NVL(d.Del_Fee, 0), 2) AS Delivery_Fee,
    ROUND(od.OD_Qty * p.Prod_Price - od.O_DISC + NVL(d.Del_Fee, 0), 2) AS
Final_Price,
    o.OD_PayMethod AS Payment_Method,
    NULL AS Total_Invoice
FROM Orders o
JOIN Customer c ON o.Cust_ID = c.Cust_ID
JOIN OrderDetail od ON o.Ord_ID = od.Ord_ID
JOIN Product p ON od.Prod_ID = p.Prod_ID
LEFT JOIN Delivery d ON o.Ord_ID = d.Ord_ID
UNION ALL
SELECT
    o.Ord_ID AS Order_ID,
    c.Cust_Name,
    c.Cust_Contact,
    TO_CHAR(o.Ord_Date, 'DD-MM-YYYY') AS Order_Date,
    NULL AS Product_Name,
    NULL AS Quantity,
    NULL AS Unit_Price,
    NULL AS Original_Price,
    NULL AS Discount_Applied,
    NULL AS Total_Price,
    NULL AS Delivery_Fee,
    NULL AS Final_Price,
    NULL AS Payment_Method,
    ROUND(SUM(od.OD_Qty * p.Prod_Price - od.O_DISC + NVL(d.Del_Fee, 0)), 2)
AS Total_Invoice
FROM Orders o
```

JOIN Customer c ON o.Cust_ID = c.Cust_ID
 JOIN OrderDetail od ON o.Ord_ID = od.Ord_ID
 JOIN Product p ON od.Prod_ID = p.Prod_ID
 LEFT JOIN Delivery d ON o.Ord_ID = d.Ord_ID
 GROUP BY o.Ord_ID, c.Cust_Name, c.Cust_Contact, o.Ord_Date
 ORDER BY Order_ID, Product_Name NULLS LAST;

ORDER_ID	CUST_NAME	CUST_CONTACT	ORDER_DATE	PRODUCT_NAME	QUANTITY	UNIT PRICE	ORIGINAL PRICE	DISCOUNT APPLIED	TOTAL PRICE	DELIVERY_FEE	FINAL PRICE	PAYMENT_METHOD	TOTAL_INVOICE
O001	Ali Baba	0123456789	01-05-2025	NASI BRIYANI GAM AYAM	2	13.9	27.8	0	27.8	15	42.8	CreditCard	81.7
O001	Ali Baba	0123456789	01-05-2025	RENDANG DENDENG (200g)	1	23.9	23.9	0	23.9	15	38.9	CreditCard	
O002	Siti Liza	0172345678	02-05-2025	2 Compartment Lunch Box (600ml)	3	285	855	0	855	20	875	BankTransfer	918
O002	Siti Liza	0172345678	02-05-2025	E20 BIO SINGLET 12x12	2	7.5	15	0	15	20	35	BankTransfer	
O003	Nurul Aini	0134567890	03-05-2025	NASI GORENG KAMPUNG	50	11.9	595	59.5	535.5	25	560.5	BankTransfer	701.6
O003	Nurul Aini	0134567890	03-05-2025	SAMBAL SOTONG KERING	10	12.9	129	12.9	116.1	25	141.1	BankTransfer	
O004	Faiz Saduk	0168765432	04-05-2025	Corn Starch 400ml bowl	20	195	3900	117	3783	18	3801	TnG	21279
O004	Faiz Saduk	0168765432	04-05-2025	Fork Corn Starch	100	180	18000	540	17460	18	17478	TnG	
O005	Jason Derulo	0145678901	05-05-2025	NASI BRIYANI GAM AYAM	15	13.9	208.5	20.85	187.65	12	199.65	CreditCard	318.75
O005	Jason Derulo	0145678901	05-05-2025	NASI GORENG UDANG	10	11.9	119	11.9	107.1	12	119.1	CreditCard	
O006	Alicia Keys	0189988776	06-05-2025	4oz corn starch sauce cup	30	225	6750	202.5	6547.5	10	6537.5	BankTransfer	7023.4
O006	Alicia Keys	0189988776	06-05-2025	E55 BIO SINGLET 22x24	20	23.5	470	14.1	455.9	10	465.9	BankTransfer	

Explanation:

This query would help the company by generating detailed invoices for customer orders including product prices, discounts, delivery fees, and payment details, along with total invoice summaries. With this, the company can have a smoother process when issuing receipts for their customers and filter them as needed. The company will be able to better manage their customers with this query.

3. Checking the Current Stock Level

```
SELECT
  p.PROD_ID,
  p.PROD_info,
  COALESCE(pur.Total_Purchased, 0) AS Total_Purchased,
  COALESCE(ord.Total_Ordered, 0) AS Total_Ordered,
  p.Prod_Qty AS Current_Stock
FROM Product p
LEFT JOIN (
  SELECT Prod_ID, SUM(P_Qty) AS Total_Purchased
  FROM PurchaseDetail
  GROUP BY Prod_ID
) pur ON p.PROD_ID = pur.Prod_ID
LEFT JOIN (
  SELECT Prod_ID, SUM(OD_QTY) AS Total_Ordered
  FROM OrderDetail
  GROUP BY Prod_ID
) ord ON p.PROD_ID = ord.Prod_ID
ORDER BY
  CASE
    WHEN p.PROD_ID LIKE 'PR%' THEN 0
    ELSE 1
  END,
  p.PROD_ID;
```

PROD_ID	PROD_INFO	TOTAL_PURCHASED	TOTAL_ORDERED	CURRENT_STOCK
PR001	NASI BRIYANI GAM AYAM	50	17	533
PR002	SAMBAL SOTONG KERING	50	10	540
PR003	RENDANG DENDENG (200G)	50	1	549
PR004	SAMBAL TUMIS IKAN BILIS	0	0	500
PR005	NASI GORENG KAMPUNG	0	50	450
PR006	NASI GORENG UDANG	0	10	490
PR007	NASI GORENG IKAN MASIN	0	0	500
PR008	NASI GORENG AYAM	0	0	500
PR009	NASI GORENG DAGING	0	0	500
PR010	KARI AYAM	0	0	500
PR011	NASI PUTIH	0	0	500
PR012	RENDANG DENDENG MINI (120G)	0	0	500
PR013	NASI BRIYANI GAM 535G	0	0	500
PR014	NASI BRIYANI HERBA 395G	0	0	500
PR015	NASI BRIYANI BUKHARI 500G	0	0	500
PR016	PES KARI KEPALA IKAN 120G	0	0	500
PR017	PES NASI GORENG KAMPUNG 120G	0	0	500
PR018	PES NASI BRIYANI GAM 120G	0	0	500
PR019	PES MASAK KERUTUK 120G	0	0	500
PR020	PES ASAM PEDAS 120G	0	0	500
PR021	PES SAMBAL TUMIS 120G	0	0	500
PR022	PES MASAK TIGA RASA 120G	0	0	500
PR023	PES MIE GORENG 120G	0	0	500
HIT-CB-001	E20 BIO SINGLET 12x12	200	2	698
HIT-CB-002	E30 BIO SINGLET 15x15	0	0	500
HIT-CB-003	E40 BIO SINGLET 17x19	0	0	500
HIT-CB-004	E48 BIO SINGLET 18x22	100	0	600
HIT-CB-005	E55 BIO SINGLET 22x24	0	20	480
HIT-CB-006	E60 BIO SINGLET 24x30	0	0	500
HIT-CB-007	PAPER BAG WITH BASE SOS NO.4	0	0	500
HIT-CB-008	PAPER BAG WITH BASE SOS NO.6	0	0	500
HIT-CB-009	PAPER BAG WITH BASE SOS NO.8	0	0	500
HIT-CTR-001	2 Compartment Lunch Box (600ml)	200	3	697
HIT-CTR-002	4oz corn starch sauce cup	0	30	470
HIT-CTR-003	2oz corn starch sauce cup	0	0	500
HIT-CTR-004	Corn Starch 800ml bowl	0	0	500
HIT-CTR-005	Corn Starch 400ml bowl	150	20	630
HIT-CTR-006	Corn Starch 450ml bowl	0	0	500
HIT-CTR-007	Corn Starch 280ml bowl	0	0	500
HIT-CTR-008	4 compartment lunch box	0	0	500
HIT-CTR-009	3 compartment lunch box	0	0	500
HIT-CUP-001	16oz Corn starch Cup	200	0	700
HIT-CUP-002	16oz Cornstarch Cold Cup Lid	150	0	650
HIT-CUP-003	90mm paper lid - white	0	0	500
HIT-CUP-004	90mm PLA lid	0	0	500
HIT-CUP-005	80 mm paper based lid white	0	0	500
HIT-CUP-006	12oz Bio Bagasse Cup (Without Lid)	0	0	500
HIT-CUP-007	12oz RIPPLE WRAP PAPER CUP	0	0	500
HIT-CUP-008	8oz RIPPLE WRAP PAPER CUP	0	0	500
HIT-CUP-009	Paper Cup 16oz	0	0	500
HIT-UTS-001	Wooden Stirrer	100	0	600
HIT-UTS-002	Spoon Corn Starch	0	0	500
HIT-UTS-003	Fork Corn Starch	0	100	400
HIT-UTS-004	Wooden Spoon	0	0	500
HIT-UTS-005	Wooden Fork	0	0	500
HIT-UTS-006	3 in 1 Cutlery	0	0	500

Explanation:

This query helps the company monitor the stock levels of each product by showing the total quantity purchased from suppliers, the total quantity ordered by customers, and the current quantity available in inventory. By analysing this data, the company can identify which products need restocking and which are performing well in terms of sales. This allows for better inventory planning, reduces the risk of overstocking or running out of popular items, and helps maintain a smooth supply operation for the company.

4. Total Purchase Revenue Grouped by Customer

```
SELECT
  C.CUST_ID,
  C.CUST_NAME,
  C.CUST_TYPE,
  SUM(OD.ORD_PRICE) AS TOTAL_SPENT
FROM
  CUSTOMER C
JOIN
  ORDERS O ON C.CUST_ID = O.CUST_ID
JOIN
  ORDERDETAIL OD ON O.ORD_ID = OD.ORD_ID
GROUP BY
  C.CUST_ID, C.CUST_NAME, C.CUST_TYPE
ORDER BY
  C.CUST_ID;
```

CUST_ID	CUST_NAME	CUST_TYPE	TOTAL_SPENT
A001	Nurul Aini	agent	651.6
A002	Faiz Saduk	agent	21243
C001	Ali Baba	customer	51.7
C002	Siti Liza	customer	870
D001	Jason Derulo	dropshipper	294.75
D002	Alicia Keys	dropshipper	7003.4

Explanation:

This query helps the company identify which customers generate the highest sales revenue and the types of products they have purchased. This can allow the company to make decisions and prioritize the satisfaction of high-value customers. The company can focus on retaining these key customers and improving their experience by analysing this data.

5. Most Ordered Products by Quantity

```

SELECT
    P.PROD_ID,
    P.PROD_INFO,
    P.PROD_TYPE,
    P.PROD_SUBCATEGORY,
    SUM(OD.OD_QTY) AS TOTAL_ORDERED,
    SUM(OD.OD_QTY * P.PROD_PRICE) AS TOTAL_SALES
FROM
    ORDERDETAIL OD
JOIN
    PRODUCT P ON OD.PROD_ID = P.PROD_ID
GROUP BY
    P.PROD_ID, P.PROD_INFO, P.PROD_TYPE, P.PROD_SUBCATEGORY
ORDER BY
    TOTAL_ORDERED DESC;

```

PROD_ID	PROD_INFO	PROD_TYPE	PROD_SUBCATEGORY	TOTAL_ORDERED	TOTAL_SALES
HIT-UTS-003	Fork Corn Starch	Biodegradable Packaging Items	Utensil	100	18000
PR005	NASI GORENG KAMPUNG	Sharifah Ready to Eat Products	Ready to Eat	50	595
HIT-CTR-002	4oz corn starch sauce cup	Biodegradable Packaging Items	Container	30	6750
HIT-CB-005	E55 BIO SINGLET 22x24	Biodegradable Packaging Items	Carry Bag	20	470
HIT-CTR-005	Corn Starch 400ml bowl	Biodegradable Packaging Items	Container	20	3900
PR001	NASI BRIYANI GAM AYAM	Sharifah Ready to Eat Products	Ready to Eat	17	236.3
PR002	SAMBAL SOTONG KERING	Sharifah Ready to Eat Products	Ready to Eat	10	129
PR006	NASI GORENG UDANG	Sharifah Ready to Eat Products	Ready to Eat	10	119
HIT-CTR-001	2 Compartment Lunch Box (600ml)	Biodegradable Packaging Items	Container	3	855
HIT-CB-001	E20 BIO SINGLET 12x12	Biodegradable Packaging Items	Carry Bag	2	15
PR003	RENDANG DENDENG (200G)	Sharifah Ready to Eat Products	Ready to Eat	1	23.9

Explanation:

This query shows which products are selling the most by quantity. The results are sorted so that the most ordered products appear at the top of the table. This helps the company determine which products are popular and make better decisions about inventory, marketing, or promotions based on the actual sales performance.

6. Unpurchased Products

```
SELECT p.Prod_ID, p.Prod_Info, p.Prod_Qty
FROM Product p
WHERE p.Prod_ID NOT IN (
    SELECT DISTINCT od.Prod_ID
    FROM OrderDetail od
)
ORDER BY
    CASE
        WHEN p.Prod_ID LIKE 'PR%' THEN 0
        ELSE 1
    END,
    p.Prod_ID;
```

PROD_ID	PROD_INFO	PROD_QTY
PR004	SAMBAL TUMIS IKAN BILIS	500
PR007	NASI GORENG IKAN MASIN	500
PR008	NASI GORENG AYAM	500
PR009	NASI GORENG DAGING	500
PR010	KARI AYAM	500
PR011	NASI PUTIH	500
PR012	RENDANG DENDENG MINI (120G)	500
PR013	NASI BRIYANI GAM 535G	500
PR014	NASI BRIYANI HERBA 395G	500
PR015	NASI BRIYANI BUKHARI 500G	500
PR016	PES KARI KEPALA IKAN 120G	500
PR017	PES NASI GORENG KAMPUNG 120G	500
PR018	PES NASI BRIYANI GAM 120G	500
PR019	PES MASAK KERUTUK 120G	500
PR020	PES ASAM PEDAS 120G	500
PR021	PES SAMBAL TUMIS 120G	500
PR022	PES MASAK TIGA RASA 120G	500
PR023	PES MIE GORENG 120G	500
HIT-CB-002	E30 BIO SINGLET 15x15	500
HIT-CB-003	E40 BIO SINGLET 17x19	500
HIT-CB-004	E48 BIO SINGLET 18x22	600
HIT-CB-006	E60 BIO SINGLET 24x30	500
HIT-CB-007	PAPER BAG WITH BASE SOS NO.4	500
HIT-CB-008	PAPER BAG WITH BASE SOS NO.6	500
HIT-CB-009	PAPER BAG WITH BASE SOS NO.8	500
HIT-CTR-003	2oz corn starch sauce cup	500
HIT-CTR-004	Corn Starch 800ml bowl	500
HIT-CTR-006	Corn Starch 450ml bowl	500
HIT-CTR-007	Corn Starch 280ml bowl	500
HIT-CTR-008	4 compartment lunch box	500
HIT-CTR-009	3 compartment lunch box	500
HIT-CUP-001	16oz Corn starch Cup	700
HIT-CUP-002	16oz Cornstarch Cold Cup Lid	650
HIT-CUP-003	90mm paper lid - white	500
HIT-CUP-004	90mm PLA lid	500
HIT-CUP-005	80 mm paper based lid white	500
HIT-CUP-006	12oz Bio Bagasse Cup (Without Lid)	500
HIT-CUP-007	12oz RIPPLE WRAP PAPER CUP	500
HIT-CUP-008	8oz RIPPLE WRAP PAPER CUP	500
HIT-CUP-009	Paper Cup 16oz	500
HIT-UTS-001	Wooden Stirrer	600
HIT-UTS-002	Spoon Corn Starch	500
HIT-UTS-004	Wooden Spoon	500
HIT-UTS-005	Wooden Fork	500
HIT-UTS-006	3 in 1 Cutlery	500

Explanation:

This query helps the company identify products that have never been ordered by any customers. By listing all products that do not appear in the order records, the company can evaluate which items are not generating sales. This insight allows the company to make an informed decision about discontinuing, promoting or repackaging these unpurchased products to improve inventory efficiency and reduce costs.